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ORIGINAL ARTICLE

Impact of cannabis on non-medical opioid use and symptoms of posttraumatic stress disorder: a nationwide longitudinal VA study

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ABSTRACT

Background: Multiple states have authorized cannabis as an opioid substitution agent and as a treatment for posttraumatic stress disorder (PTSD).
Objectives: This study sought to investigate the relationship between cannabis use, non-medical opioid use, and PTSD symptoms among U.S. veterans.
Methods: From 1992–2011, veterans admitted to specialized intensive PTSD treatment participated in a national evaluation with assessments at intake and four months after discharge. Participants with non-medical opioid use ≥ 7 days during the 30 days preceding admission were divided into two groups: those with cannabis use ≥ 7 days, and those without cannabis use. These two groups were compared on measures of substance use and PTSD symptoms at baseline and 4-months outpatient follow-up. We hypothesized that, at both assessments, the group with baseline cannabis use would show less non-medical opioid use and less severe PTSD symptoms.
Results: Of 1,413 veterans with current non-medical opioid use, 438 (30.3%) also used cannabis, and 985 (69.7%) did not. At baseline, veterans with concurrent non-medical opioid and cannabis use had slightly fewer days of non-medical opioid use ($p < .005$; d = −0.16), greater use of other substances ($p < .0001$) and more PTSD symptoms ($p = .003$; d = 0.16), compared to veterans who used non-medical opioids but not cannabis. At follow-up, substance use or PTSD symptoms did not significantly differ.
Conclusion: Cannabis use was not associated with a substantial reduction of non-medical opioid use, or either improvement or worsening of PTSD symptoms in this population. Hence, these data do not encourage cannabis use to treat either non-medical opioid use or PTSD.

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Marijuana; adverse effect; opioid-sparing effect; cannabinoid; hyperarousal

Introduction

The United States in the midst of an opioid epidemic, with over 100 Americans dying from overdoses daily, and an approximate cost of 80 billion dollars per year (1,2). The enormous societal and economic burden of this epidemic has fueled several harm reduction efforts. Notable interest has arisen regarding the clinical benefits and risks of cannabis for individuals with opioid user disorder (OUD) and Posttraumatic Stress Disorder (PTSD).

At the time of this writing, medical cannabis (MC) is legal in at least 33 states and Washington D.C (3). One of the common qualifying conditions for MC is PTSD (3,4). More recently, a number of states have modified existing MC laws to allow individuals to replace prescription opioid medications with cannabis (5). Other states have added OUD to the list of qualifying conditions, on the premise that cannabis may attenuate opioid use (5).

The notion of opioid-sparing effects of cannabinoids (i.e., reduction of the active opioid dose with the intent of curtailing their adverse effects) is dovetailed by preclinical evidence of interactions between the opioid and the endocannabinoid systems at the anatomical, functional and behavioral levels (6,7). There has also been growing interest in investigating the endocannabinoid system as a potential target for the treatment of PTSD (8,9).

Despite the preclinical findings, the approval of MC for both opioid substitution and for the treatment of PTSD has occurred in the absence of data meeting the standards required for approval by the Food and Drug Administration (10,11). Furthermore, prior observational studies of the relationship between opioid and cannabis use either have relied on aggregated data at the state level, exclusively examined medical opioid use, or had limited validity and generalizability for individuals with PTSD – as they focused on other clinical populations (e.g. cancer pain, HIV polyneuropathy) (12–14).

There are multiple proposed explanations for the high rates of concurrent opioid and cannabis use in veterans with PTSD. These include self-medication of

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anxiety, chronic pain and symptoms of opioid withdrawal, which may all overlap with symptoms of PTSD (15,16). However, the evidence for analgesic and opioid withdrawal suppression effects of cannabinoids is derived from studies that excluded individuals with PTSD (17,18). Although survey studies indicate that individuals with PTSD report subjective relief from trauma-related symptoms following cannabis use (19–22), longitudinal studies suggest cannabis use is associated with poorer PTSD treatment outcomes and a higher prevalence of cannabis use disorder (CUD) (23,24). Observational studies also indicate that specific symptoms of PTSD, such as avoidance and hyperarousal, are associated with higher frequency of cannabis use (25); and that cannabis use may increase suicidal thoughts and behavior among veterans with PTSD (26). Notably, PTSD has also been associated with higher rates of non-medical opioid use and OUD (27). Taken together, these data highlight the need to evaluate the impact of concurrent non-medical opioid use and cannabis use among individuals with PTSD.

Given the paucity of research in this area, an alternative approach to examine the effects of concurrent cannabis use on non-medical opioid use among individuals with PTSD is to utilize administrative program evaluation data from treatment-seeking patients. One source of data to address this gap is the Veterans Health Administration (VHA), which treats hundreds of thousands of patients with war-related PTSD each year. Wilkinson *et al.* showed that VHA patients who reported abstinence from cannabis 4 months following discharge from inpatient PTSD programs showed greater improvement in PTSD symptoms, in addition to less violent behavior, compared to those who initiated or continued previous cannabis use (28). To this date, however, a specific examination of the effects of cannabis on non-medical opioid use and on the outcomes of PTSD treatment has not yet been conducted. Convergent evidence suggests that cannabis carries a lower overall risk of negative clinical outcomes (including overdose, other medical complications and functional impairment) compared to opioids, both in the general population (1,29–31) and in individuals with PTSD (32).

In 1992, the VHA implemented a national data collection system that, by 2011, had monitored outcomes of over 47,000 individuals treated in specialized intensive PTSD programs – which served veterans with severe forms of PTSD (33). We use data from all sites participating in this program evaluation effort over the 20-year period. Because of the large sample size, we were able to identify subsamples with substantial non-medical opioid use prior to admission, some of whom also had substantial cannabis use. Hence, in this exploratory study, we sought to examine the relationships between cannabis use, non-medical opioid use, and PTSD symptoms (34–37). Based on previous literature suggesting that cannabis may attenuate the use and adverse consequences of opioids, we hypothesized that cannabis use would be associated with less non-medical opioid use, and thereby less severe symptoms of PTSD.

Methods

Participants

Data were drawn from the national evaluation of specialized intensive PTSD programs implemented by the Northeast Program Evaluation Center (NEPEC) of the Veterans Health Administration from 1992–2011. This national evaluation had the primary objective of identifying patterns of service delivery, clinical needs and outcomes of veterans receiving specialized intensive treatment for PTSD. The evaluation addressed PTSD treatment programs including short-term acute care admission units, specialized intensive inpatient units, day hospital programs and residential treatment facilities. All provided daily treatment for PTSD, but none specialized in treatment of military sexual trauma or substance use disorders.

All patients entering these programs were evaluated at baseline and 4 months after discharge, using a standardized set of sociodemographic and clinical measures. The sample from which the subjects were selected included 47,310 veterans from 57 distinct nationwide sites. Of these total sample, patients with non-medical opioid use ≥ 7 days during the 30 days preceding the admission were subdivided into two groups: those with cannabis use ≥ 7 days, and those without cannabis use. The cutoff of ≥ 7 days was chosen to include a sizable number of veterans with higher likelihood of clinically significant opioid and cannabis use.

Measures

Measures available from the dataset included socio-demographic, and military service–related characteristics, in addition to detailed clinical data on severity of substance use, PTSD symptoms and treatment program participation.

Socio-demographic and military service–related variables

Socio-demographic data included age, gender, race, marital and employment status, service connected-disability status, and war zone experience. The following dichotomous variables represent veterans' exposure to traumatic events: whether they participated in or witnessed what they considered atrocities; experienced sexual trauma encountered during active military duty; or noncombat nonsexual trauma (e.g. training accidents) in the course of their military duties. Past history of incarceration and recent suicidality (whether veteran attempted suicide in the past 4 months) were also documented at the time of program entry and follow-up and measured dichotomously. Baseline demographical data also included remote history of incarceration and violent behavior over the last 30 days – the latter assessed using a four-item self-report questionnaire from the National Vietnam Veterans Readjustment Study (38).

Clinical data

These data included diagnoses of alcohol abuse/dependence, drug abuse/dependence, and major psychiatric disorders other than PTSD (i.e. anxiety disorder, affective disorder, bipolar disorder, schizophrenia or personality disorders). The diagnoses were made by clinical interviews conducted by program staff.

Psychiatric History included documented responses to questions about past inpatient and outpatient psychiatric treatment, and receipt of psychotropic medications over the preceding 30 days, measured dichotomously.

Measures of Substance Use. included number of days in the previous 30 days on which patients used each of several addictive substances (opioids, cannabis, alcohol, sedatives, cocaine and amphetamines). These responses were added together to form the primary outcome variable, an index of recent active substance use, with a potential range of 1–150, consistent with prior reports (39). A second measure was created reflecting the number of distinct addictive substances that were used in the 30 days (range 1–5) (39). Additionally, at the 4-month outpatient follow-up interview the proportions of individuals reporting abstinence from all substances in the previous 30 days was calculated. Additionally, the Addiction Severity Index (ASI) was used to assess problem severity in alcohol use (six items; Cronbach's α =.84), as well as other drug use (ten items; α = .75). These are continuous measures with weighted scores ranging from 0.0 to 1.0 (average α = .74). All items are transposed to 0–1 scales or log-transformed and averaged. Previous studies have demonstrated these measures have inter-rater reliability and validity with an average concordance of .89 (40).

Symptoms of PTSD were assessed with the 14 items comprising the short form of the Mississippi Scale for PTSD (MISS) (41), and the four items that constitute the NEPEC PTSD assessment tool (42). These items were combined to form a total PTSD severity score.

Six subscales were constructed based on face value interpretation of items, representing: numbness, arousal, reexperiencing of past trauma, avoidance, suicidal thoughts, and irritability and violent behavior (43). Response options to individual items ranged from 1 = *never* to 5 = *frequently*. These subscales have been used in several other studies involving the same dataset (44–47).

Cronbach's α for the 18 items contained in the total PTSD symptom score was high (α = .81 at baseline and α = .71 at follow-up). Subscale α ranged from .41 – .72 at baseline and .52 – .73 at follow-up and were more interpretable than the results of an exploratory factor analysis of these data (47). Limited data supporting concurrent validation against other PTSD measures has previously been published (41). Likewise, the correlation of the short scale with the full Mississippi Scale was α = .95 at both intake and 4 months, and α = .96 at both 8 months and 1 year (41). We used the face value subscales for consistency with other studies from this data set and because they were more substantively interpretable than the results of a formal factor analysis.

Treatment Program Participation included length of stay (number of days), in addition to personal commitment to treatment and satisfaction with services. The latter two variables were derived from the NEPEC PTSD assessment tool and had a range of 1–5, with 1 reflecting low personal commitment and satisfaction, and 5 reflecting high personal commitment and satisfaction.

Data analysis

First, chi square tests for categorical variables and generalized linear models for continuous variables were employed to compare individuals with PTSD and non-medical opioid use, with and without concurrent cannabis use, on sociodemographic and military service-related characteristics, baseline clinical variables, and levels of program participation that could potentially confound comparison of post-discharge outcomes between the groups. Because of the large sample size and multiple comparisons, an α level of 0.01 was used to test for statistical significance.

Variables that were found to be significant on the bivariate analysis were used as covariates to adjust for potential selection biases in subsequent generalized linear models, which compared the groups on non-medical opioid use, other substance use, and PTSD symptom severity at baseline.

Table 1. Socio-demographic characteristics at baseline of 1413 U.S. combat veterans treated in VHA specialized PTSD treatment programs between 1992 and 2011, stratified by cannabis and opioid use.

	Opioids + Cannabis (n = 428; 30.3%)		Opioids (n = 985; 69.7%)			
Variable	*mean*	*SD*	*mean*	*SD*	*F(df)*	*P*
Age	45.8	10.1	48.4	9.2	24.7(1)	<.0001
Gender	*n*	*%*	*n*	*%*	*χ2(df)*	*P*
Male	320	74.7	660	67.0	0.32(1)	0.56
Race						
White	252	58.8	531	59.3	0.02(1)	0.87
Black	128	29.9	264	29.5	0.02(1)	0.87
Hispanic	25	5.8	57	6.3	0.13	0.70
Marital Status						
Married	76	17.7	294	32.85	32.7(1)	<.0001
Separated/Divorced	240	56.0	394	44.0	16.85(1)	<.0001
Widow	10	2.3	19	2.2	0.06(1)	0.80
Never married	99	23.2	278	28.2	4.84(1)	0.27
Service Connection						
Any for Posttraumatic stress disorder	141	33.2	336	37.7	2.51(1)	0.11
Any for Medical	350	37.8	550	40.9	1.87(1)	0.17
Any for Psychiatric	38	4.13	72	5.40	1.72(1)	0.19
Employment Status on Program Entry						
Not working now	387	91.0	791	88.8	2.75(2)	0.25
Part-time	6	1.41	25	2.81	2.75(2)	0.40
Full-time	32	7.5	74	8.31	2.75(2)	0.40
Military Characteristics						
Served in the War Zone	367	86.3	788	88.4	1.16(1)	0.28
Trauma History						
Received Incoming Fire	361	84.9	760	85.3	0.28(1)	0.86
Participated in Atrocities	82	19.0	153	17.2	0.70(1)	0.40
Witness Atrocities	112	26.3	220	24.6	0.42(1)	0.51
Sexual Trauma During Active Duty	19	5.2	49	6.0	0.48(1)	0.48
Noncombat Nonsexual Trauma During Military Service	50	13.7	87	10.7	2.2(1)	0.13
History of Incarceration	292	68.8	542	60.9	7.86(1)	0.005
Treatment Program Participation	*mean*	*SD*	*mean*	*SD*	*F(df)*	*P*
Length of stay in the program	49.6	44.7	47.2	37.2	1.31(1)	0.25
Personal commitment to treatment (1–5)	2.44	0.5	2.5	0.5	1.22(1)	0.26
Satisfaction with services (1–5)	3.1	0.5	3.18	0.3	0.25(1)	0.61

Abbreviations: VHA: Veterans Health Administration; PTSD: Posttraumatic stress disorder.

Analysis of these two groups at follow-up similarly compared them on measures of non-medical opioid use, other substance use, and change in PTSD symptoms net of baseline levels of these variables (to adjust for regression to the mean) as well as controlling for other potentially confounding baseline variables. The study was approved by the Institutional Review Board of the West Haven Center of the VA Connecticut Healthcare System and was given a waiver of informed consent.

Results

General sample characteristics and baseline bivariate analysis

The total sample consisted of 1,413 veterans, of whom 428 (30.3%) reported a history of ≥ 7 days of concurrent non-medical opioid and cannabis use over the 30 preceding days, and 985 (69.7%) reported a history of ≥ 7 days of non-medical opioid use but no cannabis use over the 30 preceding days. Altogether, 821 (58.1%) of the total sample reported non-medical opioid use every day.

Bivariate comparison showed that, at baseline, individuals with concurrent non-medical opioid and cannabis use were younger, less likely to be married, more likely to be separated/divorced and more likely to have a history of incarceration, compared to individuals with non-medical opioid but no cannabis use (Table 1).

With regards to substance use, individuals with cannabis use were more likely to have been diagnosed with alcohol or other drug dependence, used a higher number of substances, and consumed 3 drinks or more with greater frequency in the preceding 30 days. Although they had more days of use of specific substances in the same 30-day period (sedatives, cocaine and amphetamines), higher overall index score of substance use days, and higher ASI alcohol and drug scores, individuals with concurrent non-medical opioid and cannabis use reported statistically significantly fewer days of non-

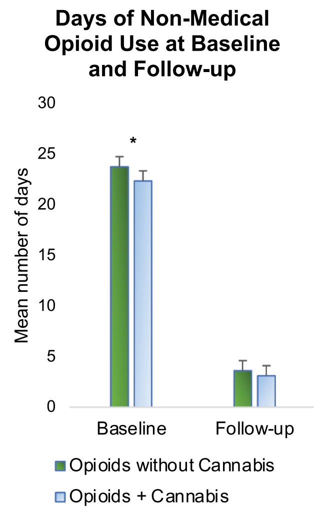

Figure 1. Days of non-medical opioid use, over the 30 days preceding each assessment, among veterans enrolled in nationwide specialized VA intensive treatment programs for posttraumatic stress disorder (PTSD). Assessments were conducted at baseline (admission) and at follow-up (4 months following discharge). At baseline, veterans with at least 7 days use in the past 30 of both cannabis use and non-medical opioid use reported statistically significantly fewer days of non-medical opioid use, compared to veterans with non-medical opioid but no cannabis use. Notably, the effect size of the reduction in non-medical opioid use associated with cannabis was small: from 23.7 ± 10.5 days to 22.3 ± 12 days (p = .005; d = −0.16). At the 4-month follow-up, there were no significant differences in days of non-medical opioid use (p = .48), between veterans with concurrent cannabis and non-medical opioid use, and veterans with non-medical opioid use but no cannabis use. The asterisk indicates a p value of less than 0.01.

medical opioid use, although the effect size was small (p = .005; d = −0.16) (Figure 1; Table 2).

With regards to PTSD symptom severity, controlling for other baseline differences, individuals with concurrent non-medical opioid and cannabis use had significantly higher total PTSD symptom scores albeit with a small effect size (p = .003; d = 0.16) (Figure 2; Table 2). The significant PTSD findings seem to have been driven primarily by higher scores in the arousal PTSD subscale (p < .001; d = 0.25). Veterans with concurrent non-medical opioid and cannabis use also reported more severe violent behavior at the time of admission (p < .001; d = 0.27).

Analysis of outcome status at 4-month follow-up

In the comparison of clinical changes between the two groups from admission to the post-discharge assessment, there were no significant differences in days of non-medical opioid use (Figure 1) or any substance use measure, including the index of substance use days controlling for baseline levels (p = .07); likewise, there were no significant differences in PTSD symptom severity, at the pre-determined alpha of 0.01 (p = .03) (Figure 2; Table 3). Abstinence from substance use at outpatient follow-up (66% of the overall sample) was also not significantly different between the two groups (p = .42). There were no differences on any measures of program participation.

Discussion

This multi-site longitudinal study examined the impact of concurrent cannabis use on non-medical opioid use and on the severity of PTSD symptoms among U.S. veterans. First, although concurrent cannabis use was associated with much greater use of other substances at baseline, individuals with concurrent cannabis use reported slightly fewer days of non-medical opioid use. Second, before program entry, we found more severe PTSD symptoms – specifically on the hyperarousal subscale – among individuals who had concurrent non-medical opioid and cannabis use. Third, at the 4-month post-discharge, outpatient follow-up assessment, we found no impact of concurrent non-medical opioid and cannabis use on days of substance use, abstinence, or PTSD symptom severity. Thus, while concurrent cannabis use was associated with slightly less non-medical opioid use in veterans with PTSD at baseline, it was accompanied by more extensive use of other substances, and more severe PTSD symptoms.

These findings are consistent with other studies that utilized individual respondent level data. In a large prospective study, cannabis use was associated with a higher prevalence of alcohol and other substance exposure,

Table 2. Clinical characteristics at baseline of 1413 U.S. combat veterans treated in VHA specialized PTSD treatment programs between 1992 and 2011, stratified by cannabis and opioid use.

	Opioids + Cannabis (n = 428; 32.3%)		Opioids (n = 985; 67.7%)			
Psychiatric History	*n*	*%*	*n*	*%*	*χ2(df)*	*P*
Attempted Suicide in the past 4 months	126	20.5	147	15.9	5.19(1)	0.02
Past Psychiatric Hospitalization	364	85.6	725	81.0	0.48(1)	0.52
Ever Outpatient Psychiatric Treatment	362	85.1	765	85.4	0.01(1)	0.96
Diagnoses and Medication Prescription						
Having Chronic Medical Diagnoses	276	64.9	630	70.7	4.58(1)	0.03
Alcohol Dependence	286	66.8	431	48.1	40.63(1)	<.0001
Drug Dependence	338	79.5	526	59.1	52.94(1)	<.0001
Anxiety Disorder	64	17.4	125	14.4	2.59(1)	0.10
Affective Disorder	126	29.7	243	27.3	0.82(1)	0.32
Bipolar Disorder	35	8.24	48	5.3	3.92(1)	0.04
Schizophrenia	7	1.6	10	1.1	0.61(1)	0.43
Personality Disorders	128	13.9	166	12.2	1.27(1)	0.25
Prescribed Psychiatric Medication (past 30 days)	69	16.2	111	12.4	3.44(1)	0.06
Clinical Status at Program Entry	*mean*	*SD*	*mean*	*SD*	*F(df)*	*P*
Substance Use						
Index of Substance Use Days (1–150)	85.8	36.3	42.1	22.3	788.0(1)	<0.001
Number of Different Substances Used	3.8	0.9	1.9	0.8	1432.9(1)	<0.001
Number of Days of ≥ 3 drinks (last 30 days)	15.4	12.6	6.1	11.2	163.04(1)	<0.001
Number of Days of Non-medical Opioid Use (last 30 days)	22.3	12.0	23.7	10.5	7.79(1)	0.005
Number of Days of Non-medical Sedative Use (last 30 days)	9.8	10.2	5.2	9.2	49.9(1)	<.0001
Number of Days of Cocaine Use (last 30 days)	11.5	12.9	4.8	9.5	135.47(1)	<.0001
Number of Days of Non-medical Amphetamine Use (last 30 days)	4.7	9.6	0.8	4.2	112.3(1)	<.0001
Number of Days of Non-medical Cannabis Use (last 30 days)	22.0	8.7	0	0	5617.79(1)	<.0001
ASI score (alcohol)	0.5	0.3	0.2	0.3	172.38(1)	<0.001
ASI score (drug)	0.4	0.1	0.2	0.1	515.7(1)	<0.001
PTSD Symptom scores						
Total PTSD symptom score	3.7	0.4	3.6	0.4	10.43(1)	0.001
Numbness	3.6	0.4	3.7	0.4	1.47(1)	0.43
Arousal	3.7	0.4	3.6	0.5	22.57(1)	<0.001
Reexperiencing	3.9	0.6	3.8	0.7	2.50(1)	0.07
Avoidance	4.1	0.8	4.1	0.9	0.18(1)	0.84
Suicidal thoughts	2.9	0.9	2.8	0.9	3.25(1)	0.07
Irritability	3.8	0.9	3.7	0.94	4.08(1)	0.03
Violent behavior (at admission)	2.4	1.40	1.9	1.3	41.14(1)	<.0001

Abbreviations: VHA: Veterans Health Administration; PTSD: Posttraumatic stress disorder; ASI: Addiction Severity Index. Covariates: Age, married, separated/divorced, history of incarceration, consuming ≥3 drinks over the last 30 days.

after adjusting for potentially confounding predictors (48). A previous cross-sectional study using nationwide VHA administrative data showed that veterans with concurrent diagnoses of OUD and CUD had a higher prevalence of other substance use disorders, and more numerous psychiatric comorbidities than those with OUD but not CUD (49). Consistent with the findings of the present study, the national VHA administrative data showed veterans diagnosed with both OUD and CUD had slightly less use of opioid use analgesics (49). Studies involving other clinical samples with opioid use, such as individuals with chronic pain, have also found that those who use cannabis have a higher prevalence of substance use disorders, without evidence of added clinical benefit in other domains (50–52).

There are several possible explanations for the specific correlates of concurrent non-medical opioid and cannabis use observed in this study of individuals with PTSD. First, some lines of evidence suggest that cannabinoids have acute anxiolytic, analgesic and opioid withdrawal suppression effects (7), which could account for a small reduction in days of opioid use (53,54). These observations may explain the negative association of cannabis and non-medical opioid use, as compared to that of cannabis and other substances. Second, patients with PTSD and concurrent non-medical opioid and cannabis use may have a distinct symptom profile (i.e., more severe hyperarousal) associated with a tendency to seek cannabis-induced anxiolysis, rather than opioid effects. Third, the polysubstance exposure found among individuals with concurrent non-medical opioid and cannabis use may decrease the likelihood of obtaining opioid prescriptions in clinical settings.

Our findings differ from those of aggregated data from policy impact studies, which report that medical and non-medical cannabis laws are associated – at the population level – with less overall risk of adverse opioid-related outcomes, such as overdose mortality (55,56),

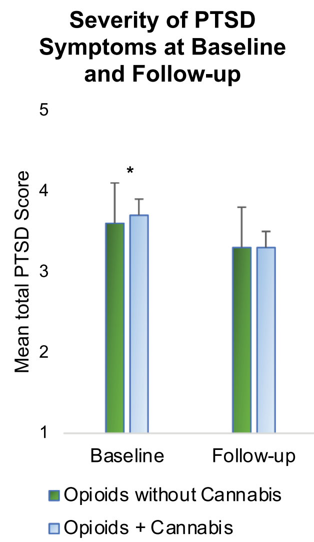

Figure 2. Posttraumatic stress disorder (PTSD) symptom severity among veterans enrolled in nationwide specialized VA intensive treatment programs for PTSD. Assessments were conducted at baseline (at admission) and at follow-up (4 months following discharge). Controlling for other baseline differences, veterans with concurrent cannabis and non-medical opioid use had slightly higher PSTD symptom scores, with a small magnitude of effect (p = .003; d = 0.16), compared to veterans with non-medical opioid use but no cannabis use. At the 4-month follow-up, there were no differences in the severity of PTSD symptoms (p = .03) between the two groups. The total PTSD severity score ranged from 1 to 5 and was derived from a combination of the Mississippi Scale for PTSD (MISS) and the VA Northeast Program Evaluation Center (NEPEC) PTSD assessment tool. The asterisk indicates a p value of less than 0.01.

Table 3. Outcome status at follow-up of 1413 U.S. combat veterans treated in VHA specialized PTSD treatment programs between 1992 and 2011, stratified by cannabis and opioid use.

	Opioids + Cannabis (n = 428; 32.3%)		Opioids (n = 985; 67.7%)			
Clinical Status at 4 months after the program	*mean*	*SD*	*mean*	*SD*	*F(df)*	*P*
Substance Use						
Index of Substance Use Days (0–150)	11.3	22.1	7.9	17.5	8.13(1)	0.07
Number of Different Substances Used	0.79	1.3	0.58	0.9	14.26(1)	0.08
Number of Days of ≥ 3 drinks (last 30 days)	2.18	6.97	2.0	5.2	6.51(1)	0.69
Number of Days of Non-medical Opioid Use (last 30 days)	3.13	4.6	3.62	5.7	11.1(1)	0.48
Number of Days of Non-medical Sedative Use (last 30 days)	0.76	10.2	1.42	9.21	0.62(1)	0.11
Number of Days of Cocaine Use (last 30 days)	0.71	4.62	4.88	0.62	6.72(1)	0.73
Number of Days of Non-medical Amphetamine Use (last 30 days)	0.09	2.1	0.14	1.3	2.05(1)	0.70
Number of Days of Non-medical Cannabis Use (last 30 days)	1.63	8.3	1.49	3.2	60.03(1)	0.88
Change in ASI score (alcohol)	−0.202	0.384	−0.202	0.304	232.09(1)	0.99
Change in ASI score (drug)	−0.234	0.227	−0.235	0.174	377.2(1)	0.91
Change in PTSD Symptom Scores						
Total PTSD symptom score	−0.355	0.56	−0.26	0.52	13.87(1)	0.03
Numbness	−0.23	0.58	−0.14	0.59	11.87(1)	0.03
Arousal	−0.404	0.6	−0.339	0.6	14.02(1)	0.15
Reexperiencing	−0.34	0.78	−0.26	0.77	6.40(1)	0.07
Avoidance	−0.26	1.1	−0.25	1.1	0.03(1)	0.83
Suicidal thoughts	−0.48	1.1	−0.38	1.0	10.45(1)	0.04
Irritability	−0.36	1.1	−0.37	1.1	0.39(1)	0.90
Violent behavior	−0.99	1.6	−1.02	1.5	12.44(1)	0.75

Abbreviations: PTSD: Posttraumatic stress disorder; ASI: Addiction Severity Index. Covariates: Age, married, separated/divorced, history of incarceration, consuming ≥3 drinks over the last 30 days.

prescription opioid use (57–60), opioid expenditures (12,13), opioid-related hospitalizations (61), and opioid positive toxicology assays among fatally injured drivers (62). This study differs from these prior reports in that we present an individual level analysis including only veterans diagnosed with PTSD. Hence, we were able to focus specifically on the relationship between concurrent cannabis use, non-medical opioid use, and symptoms of PTSD. Our findings highlight that inferences about the beneficial opioid-sparing effects of cannabis from aggregated data in policy studies may not be valid, particularly for clinical populations with PTSD who use nonmedical opioids.

Convergent evidence also suggests that chronic substance use, including cannabis, is predictive of worse outcomes among individuals with PTSD (28,63). Of note, multiple substance withdrawal states are associated with negative affect – including dysphoria, anxiety and irritability –, which overlap with PTSD symptoms, and may contribute to their worsening (64–66). Further, the shared neurocircuitry underlying substance withdrawal states and stress reactivity may account for greater hyperarousal, irritability and violence reported by the subset of individuals with PTSD who had more severe substance use in our sample (67–69).

Specific to cannabis, preclinical and human pharmacological studies (70,71) provide some biological plausibility to the hypothesis that self-medication of acute anxiety may partly explain the high rates of cannabis use among individuals with PTSD (72), as well as the common stress/anxiety symptom-coping motives for cannabis use noted in this population (73). However, while cannabis may *acutely* reduce anxiety, especially at low doses/potencies, *chronic* cannabis use may cause downregulation of cannabinoid 1 receptors in the brain (74,75), increasing the neuroendocrine response to stress (76), and impairing processes that are relevant to the phenomenology of PTSD, such as fear extinction (77,78).

This study is the first, to our knowledge, to examine the relationship between concurrent cannabis use, nonmedical opioid use, and PTSD symptom severity. Its strengths include a relatively large multi-site sample, a longitudinal design, and the naturalistic measurements of substance exposure and PTSD symptomatology.

Our study also has important limitations. First, we have relied on self-report data on substance use. Second, our study lacked data on the specific dose and type of opioids used (i.e. heroin vs. prescription opioids); on concomitant chronic pain; and on the potency of the cannabis used. Third, it is worth noting that the data used in this study was collected from 1992 to 2011 – before high-potency synthetic opioids became widely available. Similarly, opioid prescribing practices have changed significantly in the last decade. For instance, clinicians currently dispense smaller opioid supplies and use prescription monitoring systems to supervise long-term opioid therapy. It is possible that such changes would have led to different results if the study was replicated currently. Fourth, the results should be interpreted in light of a changing legal landscape of recreational cannabis across distinct U.S. states over the study period, which may have influenced access to cannabis. Fifth, program differences and the availability or quality of post-intervention services may have differed across study sites. However, it is notable that referrals to each specialized intensive PTSD program were made from several states, and were not based on the local population. Furthermore, we repeated all analyses using linear mixed models that account for the correlation of observations within treatment site with random effects for site. Our results remained unchanged by these analyses. Hence, it unlikely that clustering effects influencing the study outcomes. Finally, participants were predominantly male, and all U.S. military veterans PTSD requiring specialized intensive treatment, limiting generalizability to other populations. Finally, confounding effects of other types of substance use could also have affected the results, and the diagnoses were made in a real-world clinical context, which may limit their conformance to formal diagnostic criteria.

In sum, our findings should sound another note of caution in the clinical and policy debates over the legalization of cannabis for opioid substitution and for the treatment of PTSD. Future studies should characterize the precise temporal relationships between cannabis use, opioid withdrawal, and PTSD symptoms, using more precise Ecological Momentary Assessment methods (79). Finally, controlled experimental studies are also needed to evaluate the specific effects of cannabinoids with lower abuse liability (e.g. oral cannabinoid agonists, cannabidiol, and oromucosal nabiximols) on opioid-related outcomes, as well as their potential risks and benefits for individuals with PTSD.

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Additional information

The VA New England Mental Illness Research and Education Center, West Haven, CT manages The Intensive PTSD VA

Treatment database, which is stored in secure VA servers and is not available to the public. Queries can be directed to Robert Rosenheck, M.D. (robert.rosenheck@yale.edu).

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